

Image classification using color, texture and regions

Ya-Chun Cheng, Shu-Yuan Chen*

Department of Computer Science and Engineering, Yuan Ze University, 135 Yuan-Tung Rd., Nei-Li, Chung-Li, Taoyuan, 32026, Taiwan, R.O.C.

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Abstract

A new classification method using color, texture and regions is proposed in this study. Image-based features related to color and local edge patterns are used to prune irrelevant database images for each query image. The proposed region matching is then applied to find the match to the query image from among the set of candidate images in the database. The dissimilarity of each pair of images can be calculated on the basis of the matching results. Finally, all the database images in the candidate set can be sorted by ascending dissimilarity values. To achieve the classification goal, the k-NN rule is used to assign a class label to the query image. Note that the main contribution of this paper is to select proper features for representing color, texture and region, which, in turn, are used to achieve effective classification results. More important, all features used in the proposed method, no matter color or texture, are presented in the simple form of histogram, yet leading to effective results. Even in the stage of region matching, color and texture features in histograms are also used to obtain homogeneous regions and to measure dissimilarity. In addition, the proposed classification method can be applied to all kinds of color image databases rather than specific databases. The number of classes can be as versatile as required by the application. The effectiveness and practicability of the proposed method has been demonstrated by various experiments.

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1. Introduction

With the increasing popularity of using large-volume digital libraries, efficient content-based retrieval systems have become a major research and development topic. However, such systems are not easy to develop because real-world images usually contain very complex objects which make the design of similarity measures difficult. This study presents a method to unify color and texture features to obtain homogeneous regions that are, in turn, used to facilitate image retrieval. A new classification method using color, texture and regions is proposed. For image-based retrieval systems, the most common approaches use low-level features such as color, texture and shape. In earlier proposals, a single feature, such as color, is adopted to query images. Color can provide powerful information about image content. Swain and Ballard [1] proposed a color indexing algorithm to

identify color images. The 3D histograms are generated for the input and target images in the database. Images are then matched using the histogram intersection method. The method is very simple but provides good performance. Later proposals use multiple features, such as color, texture and shape, to develop a general image retrieval system. QBIC [2], Photobook [3], Virage [4], VisualSEEK [5], Netra [6] and MARS [7] are examples of this application.

On the other hand, some approaches focus on image classification. Like the image retrieval problem, classification approaches may use low-level features only [8–10] or incorporate region information [11,12] to achieve a classification. Szummer et al. [8] developed a method for inferring high-level scene properties, such as whether an image is indoor or outdoor, by classifying low-level features such as color and texture. Vailaya et al. [9,10] grouped images into semantic categories using low-level features. They combine multiple two-class classifiers into a single hierarchical classifier for classifying vacation pictures. Additionally, Smith et al. [11] presented a method for classifying and querying images based on

* Corresponding author. Tel.: +886-3-463-8800; Ext.357; Fax: +886-3-463-8850.

E-mail address: cschen@saturn.yzu.edu.tw (S.-Y. Chen).

the spatial orderings of regions or objects using composite region template. Lo et al. [12] uses image-based and cluster-based features for image classification. The former includes color-width, region ratio, edge ratio, edge density variance and initial color histogram. The latter includes pixel occurrence and region occurrence. Note that all these classification methods can be divided into two categories according to whether the query image is classified into specific classes [8–10] or general classes [11,12]. Actually, the latter is more practical for image classification application. The reasons are listed below. The general approaches can be applied to all kinds of color image databases rather than specific databases. In addition, the number of classes can be as versatile as required by application in the general case. The proposed method belongs to the general case.

Recently, images are segmented into several regions. The spatial and region information in images are then used to query and classify images. Most methods used only color features [2–5,7,11–15] rather than combined color and texture features [6,16,17] to segment images into homogeneous regions, leading to unsatisfactory retrieval or classification results.

A new classification method using color, texture and regions is proposed in this study. We adopt a color texture segmentation method that unifies color and texture features to obtain homogeneous regions. Image-based features related to color and local edge patterns are used to prune irrelevant database images for each query image. The proposed region matching is then applied to each pair of query image and database image in the candidate set. The dissimilarity of each pair can be calculated on the basis of the matching results. Finally, the database images in the candidate set can be sorted by ascending dissimilarity values. The k -NN rule is then used to assign a class label to the query image. Note that all features used in the proposed method, no matter color or texture, are presented in the simple form of histogram, yet leading to effective results. Even in the stage of region matching, color and texture features in histograms are also used to obtain homogeneous regions and to measure dissimilarity. Thus, the main contribution of this paper is to select proper features for representing color, texture and region, which, in turn, are used to achieve effective classification results. In addition, the proposed classification method can be applied to all kinds of color image databases rather than specific databases. The number of classes can be as versatile as required by the application.

In Section 2, we describe the basic concept of the image segmentation method. In Section 3, we describe how to extract features, how to derive dissimilarity measures, and how to classify the query images. Section 4 introduces two databases used in our experiment. Then, we evaluate the performance of the proposed method and compare our method to two existing methods. Finally, Section 5 presents our conclusions and describes future work.

2. Basic concepts of image segmentation

To completely represent image content, an image must be segmented into several different regions. Useful features are then collected from each region, which, in turn are used to classify the image. In this study we adopted the method proposed in Ref. [17] to roughly segment images into several homogeneous textured regions. The characteristics of each segmented region are then recorded for later classification. They include region texture descriptors such as color and edge histograms, color mean of a region, ratio of edge pixels in a region, coordinates of a region center and number of colors used in a region.

2.1. Texture descriptors

Two kinds of texture descriptors are adopted here: color histogram and local-edge pattern (LEP) histogram. The former is related to the distribution of colors in a region, while the latter, to the distribution of local edge patterns in a region. The adopted color quantization is based on the RGB color space. Each of the R , G and B bands is divided into four bins covering the range of 0–255, resulting in a color histogram of $4 \times 4 \times 4 = 64$ bins. The color histogram hc for a region R is computed using the following equation:

$$hc_m = \frac{n_m}{N}, \quad m = 0, \dots, 63, \quad (1)$$

where n_m is the number of pixels with color label m , and N is the number of total pixels in R .

The LEP originates with the texture descriptor LBP [18]. The LBP describes the spatial structure of a local texture. The 3×3 neighborhood of a pixel (Fig. 1(a)) is threshold by the value of the center pixel. The values of the pixels in the threshold neighborhood (Fig. 1(b)) are multiplied by the corresponding binomial weights shown in Fig. 1(c). The resulting values (Fig. 1(d)) are summed to obtain the LBP value.

To compute the LEP histogram, the edge image corresponding to the original image must be obtained first. The edge image is obtained by applying the Sobel edge detector to the Y -component (visual brightness) of the original image. The Y component for each pixel is

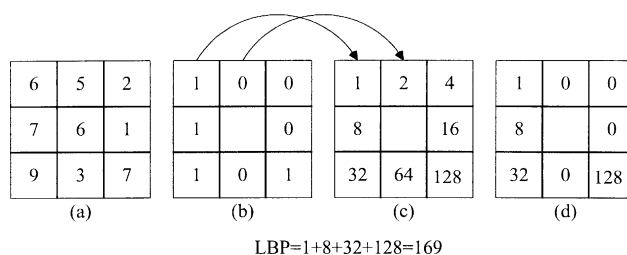


Fig. 1. Example of LBP descriptor.

computed by

$$Y = 0.299 \times R + 0.587 \times G + 0.144 \times B, \quad (2)$$

where RGB correspond to the color values. The edge image includes all the edge pixels with edge values greater than 100. From the edge image, the LEP descriptor is easy to compute. The LEP can describe the spatial structure of a local texture according to the organization of the edge pixels. For an example of edge image as shown in Fig. 2(a), the binary edge values are multiplied by the corresponding binomial weights (Fig. 2(b)). The resulting values (Fig. 2(c)) are summed to obtain the LEP value.

Finally, the LEP histogram he for a region R is computed from

$$he_m = \frac{n_m}{N}, m = 0, \dots, 511, \quad (3)$$

where n_m is the number of pixels with LEP value m , and N is the number of total pixels in R .

2.2. Homogeneous measure based on texture descriptors

The distributions of color and LEP are used to derive a homogeneity measure for segmentation. The histogram intersection technique [1] is used to compare texture descriptors. The homogeneity measure Hc of two color textured regions R_i and R'_i in the color histogram is obtained by

$$Hc = \sum_{m=0}^{63} \min(hc_m, hc'_m), \quad (4)$$

where hc_m and hc'_m are the normalized frequencies in bin m of the color histograms of regions R and R' , respectively. Note that as Hc approaches 1, the more alike the color histograms.

The homogeneity measure He of R_i and R'_i in the LEP histogram is given by

$$He = \sum_{m=0}^{511} \min(he_m, he'_m), \quad (5)$$

Note that the closer He is to 1, the more alike the LEP histograms.

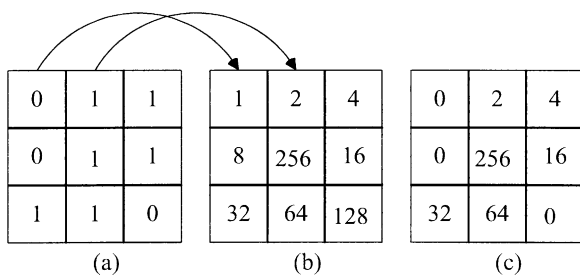


Fig. 2. Example of LEP descriptor.

Finally, the homogeneity measure H of R and R' is computed as the weighted average of Hc and He , i.e.

$$H = 0.5 \times Hc + 0.5 \times He. \quad (6)$$

2.3. Image segmentation

The segmentation algorithm proposed in Ref. [17,18] is modified to use in this study. The algorithm consists of two parts: hierarchical splitting and agglomerative merging. First, hierarchical splitting is used to divide the image into regions of roughly uniform texture. Then an agglomerative procedure is used to merge similar adjacent regions until a stopping criterion is met. Eq. (6) derived in Section 2.2 is used in this section to measure the homogeneity of a pair of textured regions.

The hierarchical splitting algorithm recursively splits the original image into four rectangular blocks of varying size. The decision whether or not to split a block is based on a homogeneity test. We use Eq. (6) to measure the six pairwise homogeneities of the four subblocks. The largest and smallest of the six H values are denoted by H_{max} and H_{min} , respectively. The block is declared to be inhomogeneous, and thus split into four subblocks, when the relative dissimilarity within the block is greater than a threshold, i.e.

$$R = \frac{H_{max}}{H_{min}} > X \quad (7)$$

The threshold X was set to 1.02 by experimentation.

Once the image has been split into blocks of roughly homogeneous texture, we apply an agglomerative procedure to merge similar adjacent regions until one of two stopping criteria is satisfied. At each stage, we merge the pair of adjacent regions which have the largest merger importance (MI) value, where MI computed from

$$MI = \frac{1}{\sqrt{p}} \times H, \quad (8)$$

where p is the number of pixels in the smaller of the two regions, and H is the homogeneity of pair of regions as calculated by Eq. (6). Once the pair of adjacent regions with the largest MI value has been found, they are merged, and their respective color and LEP histograms are summed to form the histograms of the newly merged image region. Merging proceeds until either of the following stopping rules is true:

$$\begin{cases} MIR1 = \frac{MI_{cur}}{MI_{min}} < Y, \\ MIR2 = \frac{MI_{cur}}{MI_{max}} < Z. \end{cases} \quad (9)$$

Here, MI_{min} is the smallest of all the preceding merger importance values, MI_{max} is the largest, and MI_{cur} is

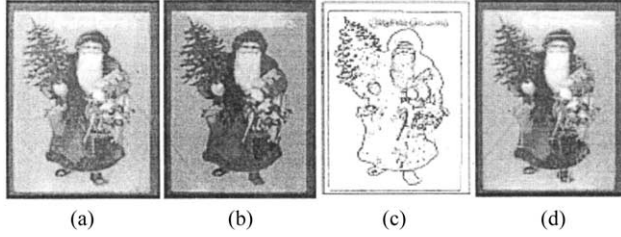


Fig. 3. Segmentation result. (a) Original image; (b) quantized image; (c) edge image; (d) segmented image.

the merger importance for the current best merge. The values of thresholds Y and Z were determined experimentally to be 0.6 and 0.12, respectively. A segmentation result is shown in Fig. 3.

3. Features and image classification

In this section we introduce how to extract features, how to derive dissimilarity measures, and how to classify the query images. Features may be extracted from original images, quantized images or segmented images. Features are categorized into image-based or region-based. The former includes the color histogram and the LEP histogram of the entire image. The latter includes the region color histogram, region LEP histogram, color mean, edge ratio, center position and color cardinality. Image-based and region-based dissimilarities are described in Sections 3.1 and 3.2, respectively. Each feature is first described, followed by the definition of dissimilarity measure based on the feature. Finally, classification strategy is included in Section 3.3.

3.1. Image-based dissimilarity

Image-based features, including color and LEP histograms, are extracted directly from images.

3.1.1. Color histogram

The color histogram is extracted from a quantized RGB image. Each of the color bands is divided uniformly into M bins covering the range of 0–255 leading to a histogram of $M \times M \times M$ bins. The proper number of bins is determined experimentally in Section 4. Histogram difference is used to measure the dissimilarity of a pair of color images. The dissimilarity of query image Q and database image T using the color histogram is calculated from

$$d_{hc}(Q, T) = \sum_{m=0}^{M \times M \times M - 1} |hc_m^{(Q)} - hc_m^{(T)}|, \quad (10)$$

where $hc_m^{(Q)}$ and $hc_m^{(T)}$ are the normalized frequencies in bin m of the color histograms.

3.1.2. LEP histogram

The dissimilarity of a pair of color images can also be measured by the difference of the LEP histograms of Eq. (3). For a query image Q and a database image T the dissimilarity in LEP histogram is calculated from

$$d_{he}(Q, T) = \sum_{m=0}^{511} |he_m^{(Q)} - he_m^{(T)}|, \quad (11)$$

where $he_m^{(Q)}$ and $he_m^{(T)}$ are the normalized frequencies in bin m of the LEP histograms.

Like the coherence vector feature [9,10,13], we divide the LEP histogram into two parts according to whether or not the center pixel is an edge point. We then have two conditional LEP histograms, i.e. edge LEP and non-edge LEP histograms, $he^{(e)}$ and $he^{(ne)}$, computed by

$$he_m^{(e)} = \frac{n_{m+256}}{N^{(e)}}, \quad m = 0, \dots, 255, \quad (12)$$

$$he_m^{(ne)} = \frac{n_m}{N - N^{(e)}}, \quad m = 0, \dots, 255,$$

where n_m is the number of pixels with LEP value m , and $N^{(e)}$ is the number of total edge pixels in R . Hereafter, we refer to Eqs. (3) and (12) as the original and conditional LEP histograms, respectively.

The dissimilarity of query image Q and database image T in the conditional LEP histogram can be calculated by the following equations:

$$d_{he}^{(c)}(Q, T) = \frac{1}{2} \{d_{he}^{(e)}(Q, T) + d_{he}^{(ne)}(Q, T)\}, \quad (13)$$

$$d_{he}^{(e)}(Q, T) = \sum_{m=0}^{255} |he_m^{(e)(Q)} - he_m^{(e)(T)}|,$$

$$d_{he}^{(ne)}(Q, T) = \sum_{m=0}^{255} |he_m^{(ne)(Q)} - he_m^{(ne)(T)}|.$$

The example in Fig. 4 shows that the conditional LEP histogram, specified by Eq. (13), has more effective discriminating power than the original LEP histogram, specified by Eq. (11). Experiments were also conducted in Section 4 to certify this claim.

3.1.3. Integration of dissimilarities

The dissimilarity of query image Q and database image T is computed by the following equation:

$$D_{\text{image}}(Q, T) = \frac{w_{hc}d_{hc}(Q, T) + w_{he}d_{he}^{(c)}(Q, T)}{w_{hc} + w_{he}} \quad (14)$$

where (w_{hc}, w_{he}) are weights assigned to the respective features. In our experiments we set all weights to 1.

3.2. Region-based dissimilarity

Since images are segmented into different numbers of regions, it is necessary to find the closest matching regions between two images so that the dissimilarity of a pair of

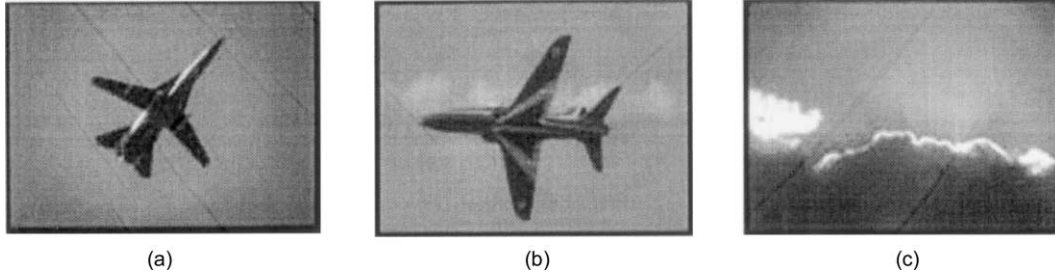


Fig. 4. Dissimilarities in different LEP histograms. (a), (b) and (c) are three images with (a) and (b) belonging to the same class, but (c) belonging to the other class. $d_{he}(a, b) = 0.0182$, $d_{he}^{(e)}(a, b) = 0.0080$, $d_{he}^{(ne)}(a, b) = 0.0965$, $d_{he}^{(e)}(a, b) = 0.0523$. $d_{he}(a, c) = 0.0164$, $d_{he}^{(e)}(a, c) = 0.0086$, $d_{he}^{(ne)}(a, c) = 0.1456$, $d_{he}^{(e)}(a, c) = 0.0771$.

images can be evaluated. Some region-based features are introduced, which are then used for region matching and derivation of region-based dissimilarity.

Region-based features are extracted directly from segmented regions. The features used in this study include color histogram, LEP histogram, color mean, edge ratio, center position and color cardinality. They are described first, followed by the integration of dissimilarities through region matching. In the following discussion, R_i is a textured region of query image Q and R_j is a textured region of database image T .

3.2.1. Region color histogram

The color histogram for a region is obtained by Eq. (1). This feature differs from an image color histogram on two counts. First, the former concerns local regions while the latter concerns the whole image. Second, the region color histogram is reduced from $M \times M \times M$ bins to 64 bins. In other words, frequencies belonging to every $(M/4)^3$ bins are combined.

The dissimilarity of the histograms of regions R_i and R_j is also computed by the following histogram difference:

$$d_{hc}^{(R)}(Q_i, T_j) = \sum_{m=0}^{63} |hc_{i,m}^{(Q)} - hc_{j,m}^{(T)}|, \quad (15)$$

where $hc_{i,m}^{(Q)}$ and $hc_{j,m}^{(T)}$ are the normalized frequencies in bin m of region color histograms for regions R_i and R_j , respectively.

3.2.2. Region LEP histogram

Like the region color histogram, region edge and non-edge LEP histograms are obtained by Eq. (12), except that the number of bins is reduced from 256 to 32. In other words, frequencies belonging to every 8 bins are combined. The dissimilarity of regions R_i and R_j in the region LEP histogram is computed by the following histogram difference:

$$d_{he}^{(R)}(Q_i, T_j) = \frac{1}{2} \left\{ \sum_{m=0}^{32} |he_{i,m}^{(e)(Q)} - he_{j,m}^{(e)(T)}| + |he_{i,m}^{(ne)(Q)} - he_{j,m}^{(ne)(T)}| \right\}, \quad (16)$$

where $he_{i,m}^{(e)(Q)}$, $he_{i,m}^{(ne)(Q)}$ and $he_{j,m}^{(e)(T)}$, $he_{j,m}^{(ne)(T)}$ are the normalized frequencies in bin m of the region LEP histograms for R_i and R_j , respectively.

3.2.3. Region color mean

Region color means, $(DC_{i,R}, DC_{i,G}, DC_{i,B})$ are the (R, G, B) mean values for region R_i and can be calculated by the following equation:

$$DC_{i,R} = \sum_{\forall \text{pixel} \in R_i} \left\{ \frac{R}{N_i} \right\}, \quad DC_{i,G} = \sum_{\forall \text{pixel} \in R_i} \left\{ \frac{G}{N_i} \right\},$$

$$DC_{i,B} = \sum_{\forall \text{pixel} \in R_i} \left\{ \frac{B}{N_i} \right\}, \quad (17)$$

where N_i is the number of total pixels belonging to region R_i .

The dissimilarity of regions R_i and R_j in the region color mean is computed by

$$d_{DC}^{(R)}(Q_i, T_j) = \frac{1}{3} \times \left\{ |DC_{i,R}^{(Q)} - DC_{j,R}^{(T)}| + |DC_{i,G}^{(Q)} - DC_{j,G}^{(T)}| + |DC_{i,B}^{(Q)} - DC_{j,B}^{(T)}| \right\}. \quad (18)$$

3.2.4. Region edge ratio

The region edge ratio represents the ratio of edge pixels in region R_i and can be calculated by the following equation:

$$ER_i = \frac{\sum_{\forall \text{edgepixel} \in R_i} 1}{N_i}. \quad (19)$$

The dissimilarity of regions R_i and R_j in the region edge ratio is

$$d_{er}^{(R)}(Q_i, T_j) = |ER_i^{(Q)} - ER_j^{(T)}|. \quad (20)$$

3.2.5. Region center position

The coordinates of region center for region R_i are calculated as follows:

$$X_i = \frac{1}{\text{ImageWidth}} \times \left\{ \frac{\sum_{\forall \text{pixel} \in R_i} x}{N_i} \right\}, \quad Y_i = \frac{1}{\text{ImageHeight}} \times \left\{ \frac{\sum_{\forall \text{pixel} \in R_i} y}{N_i} \right\}, \quad (21)$$

where (x, y) are the coordinates of each pixel in R_i .

The dissimilarity of regions R_i and R_j in the region center position is computed as

$$d_{\text{center}}^{(R)}(Q_i, T_j) = \frac{1}{2} \times \{ |X_i^{(Q)} - X_j^{(T)}| + |Y_i^{(Q)} - Y_j^{(T)}| \}. \quad (22)$$

$$d_{\text{match}}^{(R)}(Q_i, T_j) = \frac{\{w_{DC}^{(MR)} \times d_{DC}^{(R)}(Q_i, T_j) + w_{er}^{(MR)} \times d_{er}^{(R)}(Q_i, T_j) + w_{\text{center}}^{(MR)} \times d_{\text{center}}^{(R)}(Q_i, T_j) + w_{\text{color}}^{(MR)} \times d_{\text{color}}^{(R)}(Q_i, T_j)\}}{(w_{DC}^{(MR)} + w_{er}^{(MR)} + w_{\text{center}}^{(MR)} + w_{\text{color}}^{(MR)})}, \quad (25)$$

3.2.6. Region color cardinality

A color set gives a group of colors, while, a color histogram denotes the relative amounts of the colors. However, a color set can also be obtained by thresholding a color histogram. For example, each color label with its color histogram value above a threshold (1% of the region size in this study) can be viewed as a valid color. The color cardinality for region R_i can then be computed by

$$Cd_i = \sum_{m=0}^{63} C_{i,m}, \quad C_{i,m} = \begin{cases} 1, & \text{if } hc_{i,m} \geq \tau_m, \\ 0, & \text{otherwise,} \end{cases} \quad (23)$$

where $hc_{i,m}$ is the normalized frequency in bin m of the color histogram of region R_i and $\tau_m = 0.01$.

The dissimilarity of region R_i and R_j in the region color cardinality is computed by the following equation:

$$d_{\text{color}}^{(R)}(Q_i, T_j) = \frac{|Cd_i^{(Q)} - Cd_j^{(T)}|}{\max(Cd_i^{(Q)}, Cd_j^{(T)})}. \quad (24)$$

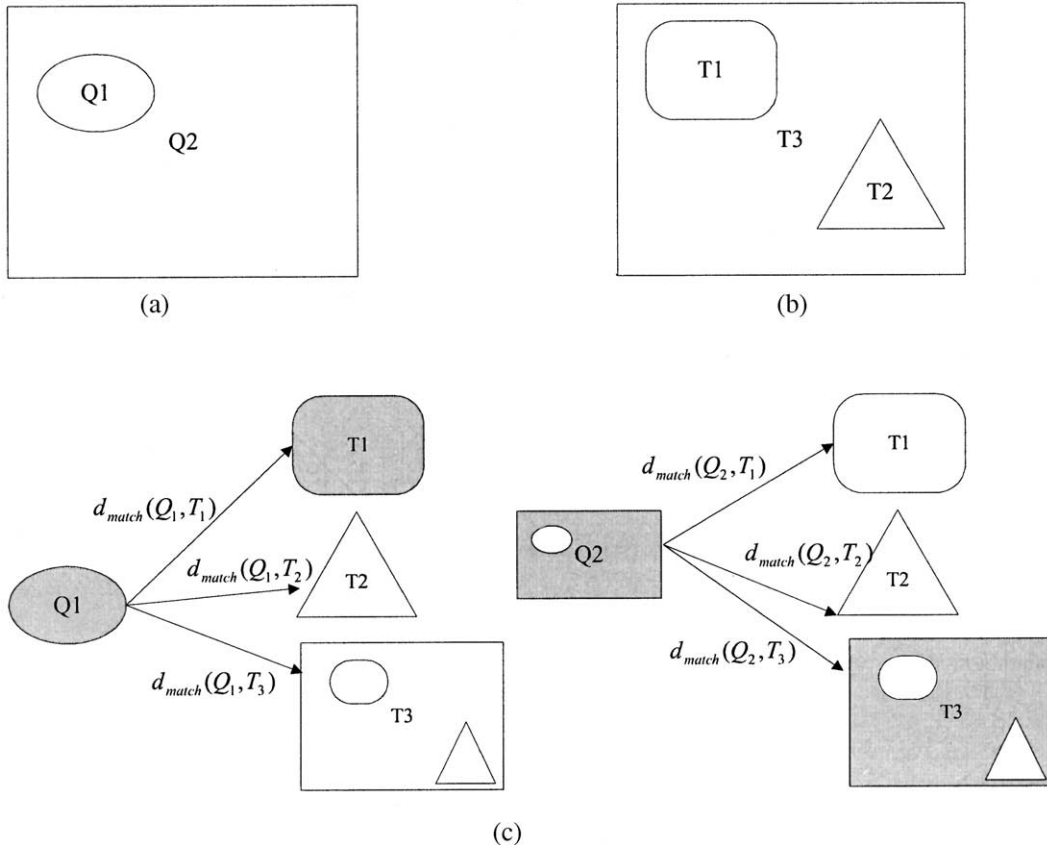


Fig. 5. Concept of region matching. (a) Query image; (b) database image; (c) region matching.

3.2.7. Integration of dissimilarities

In this section, the matching algorithm based on region features is introduced first, followed by the derivation of region-based dissimilarity for each pair of images.

The matching score of regions R_i and R_j is calculated by Eq. (25), where $(w_{DC}^{(MR)}, w_{er}^{(MR)}, w_{center}^{(MR)}, w_{color}^{(MR)})$ are weights assigned to the respective four features. In our experiments we set all weights to 1.

The matched region for R_i , denoted by $R_{mp(i)}$ is then chosen as the region with the smallest score, i.e.

$$mp(i) = \arg \min_{R_j \in T} d_{match}^{(R)}(Q_i, T_j). \quad (26)$$

Note that the matching algorithm allows duplicates. That is, it is possible that two regions in Q match the same region in T . Fig. 5 illustrate the concept of region matching. An example of region matching is shown in Fig. 6.

The dissimilarity of query image region R_i and its matched region $R_{mp(i)}$ in the database image T is calculated by

$$d_{region}(Q_i, T_{mp(i)}) = \frac{w_{hc}^{(DR)} \times d_{hc}^{(R)}(Q_i, T_{mp(i)}) + w_{he}^{(DR)} \times d_{he}^{(R)}(Q_i, T_{mp(i)}) + w_{er}^{(DR)} \times d_{er}^{(R)}(Q_i, T_{mp(i)}) + w_{color}^{(DR)} \times d_{color}^{(R)}(Q_i, T_{mp(i)})}{(w_{hc}^{(DR)} + w_{he}^{(DR)} + w_{er}^{(DR)} + w_{color}^{(DR)})}, \quad (27)$$

where $(w_{hc}^{(DR)}, w_{he}^{(DR)}, w_{er}^{(DR)}, w_{color}^{(DR)})$ are the feature weights. In our experiments we set all weights to 1.

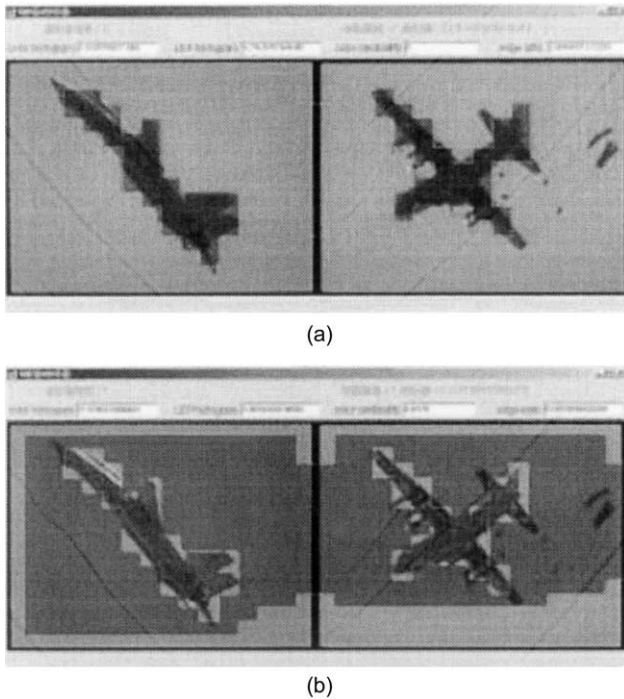


Fig. 6. Example of region matching. (a) and (b) two matching pairs.

Region-based dissimilarity of query image Q and database image T is then computed by

$$D_{region}(Q, T) = \frac{\sum_{i=1}^{NR^{(Q)}} d_{region}(Q_i, T_{mp(i)})}{NR^{(Q)}}, \quad (28)$$

where $NR^{(Q)}$ is the number of regions in the query image Q .

3.3. Classification strategy

Since region matching is time expensive, the proposed classification method adopts a filtering strategy to prune irrelevant images in advance. The region-based features are then extracted from only a small set of candidate images. Therefore, the processing time is significantly reduced.

In the filtering stage, the image-based dissimilarity $D_{image}(Q, T)$, specified by Eq. (14), is used to evaluate the dissimilarity of query image Q and database image T . Those database images with high dissimilarity values are then pruned. In our experiments 39 out of 40 database

images get pruned if the number of total database images is not less than 1600; otherwise 40 database images remain. Some experiments are conducted in Section 4 to show that the pruning rate is appropriate.

After filtering has been completed, we apply region matching to each pair of query image and database image from the candidate set. The region-based dissimilarity $D_{region}(Q, T)$ for the pair is then calculated by Eq. (28). The final dissimilarity is obtained from the following equation:

$$D(Q, T) = D_{image}(Q, T) + D_{region}(Q, T) \quad (29)$$

All database images in the candidate set can be sorted by ascending dissimilarity values. The k -nearest-neighbor rule (k -NN rule) is then used to classify the query image. The first k images are used to determine the class of the query image. The query image is assigned to the class majority among the first k images. On the other hand, the rejection version of the k -NN rule is designed as follows. If greater than $k/2$ images of the first k images have the same class, the query image is assigned to this class. Otherwise, the query image is rejected and not assigned to a class.

4. Experimental results

Two databases are used in our experiments. Both are collected from the Corel stock photo library. The first, called



Fig. 7. Representative images from Corel-20.

Corel-20, includes 20 classes; the other, called Corel-6, includes 6 classes. Each class includes 80 database images and 20 test images. Hence, there are 1,600 images in database and 400 test images for Corel-20, and 480 database images and 120 test images for Corel-6. Representative images from Corel-20 and Corel-6 are shown in Figs. 7 and 8, respectively. Note that the classes of Corel-6 are the same as those in Ref. [10], i.e. City, Sunset, Beach, Farm, Mountain and Women.

The proposed method has been implemented on an IBM compatible PC with a single Intel Pentium III 800 MHz CPU and 384 megabytes SDRAM. The operating system is Microsoft Windows 2000 Server Chinese version Service Pack 2. The program was developed in the C++ language and compiled under Borland C++ Builder version 4.0. PhotoShop Chinese version 6.0.1 was used to reduce image size. The reduction criteria are, first, the width to height aspect ratio for each image must be retained, and second, the maximum value of width or height must be fixed at 512 pixels.

The experiments were designed to classify each query image into a single class. Because all the query images are

from the Corel database, their true classes are known. For brevity only the classification accuracies for the 1-NN rule, 13-NN rule and 13-NN rule with rejection are discussed. Classification accuracy for a set of images is computed as the ratio of the number of images correctly classified using the k -NN rule to the number of total query images in the set.

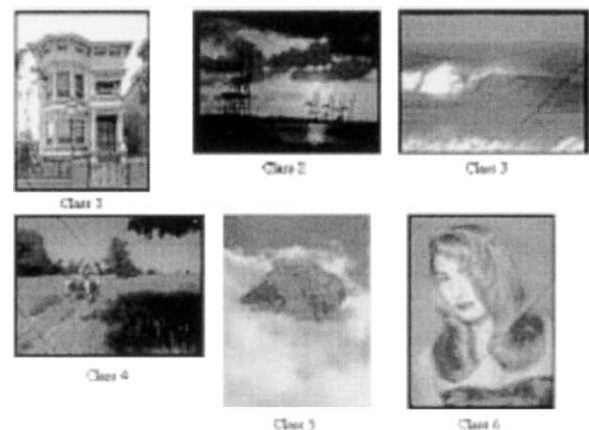


Fig. 8. Representative images from Corel-6.

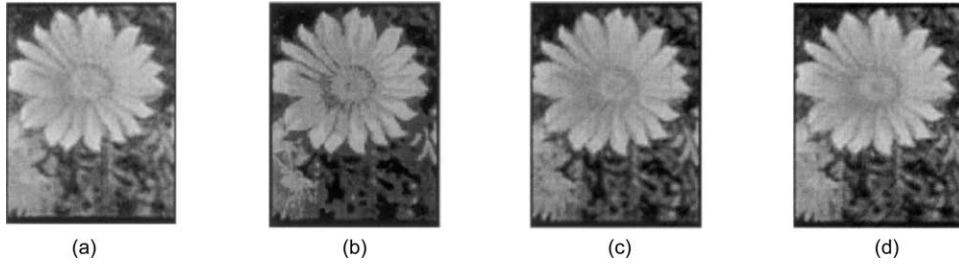


Fig. 9. Different quantization results. (a) Original image; (b) quantized image with 4 bins; (c) quantized image with 8 bins; (d) quantized image with 16 bins.

On the other hand, classification accuracy with rejection for a set of images is computed as the ratio of the number of images correctly classified using k -NN rule to the number of query images in the set, provided that these query images are not rejected. Hereafter, we choose 1-NN rule for classification, except when specially noted.

Experiments were conducted on the two databases. For each database, the classification method is applied to the sets of test and database images, respectively. In addition, a leave-one-out strategy is adopted for the latter set.

4.1. Determination of parameters

Many options and parameters need to be chosen in the proposed method. For example, the number of bins in the color histogram, the source of edge images, the type of LEP histogram, the combination of different features, etc. The options and parameters are systematically determined according to the experimental results on Corel-20 in this section.

4.1.1. Options related to the color histogram

Although other color spaces are preferred in several studies [5,11,16,17], the RGB color space is chosen in our experiment since it is not necessary to transform it into another space, which, in turn, leads to faster response time for query.

Two options related to the color histogram must be decided. First, how many bins are set for each band. Second, how is to combine the three histograms, each corresponding to one of the RGB bands. The number of bins is set to 4, 8, or 16. The combination of RGB bands

can be conjunction ($\times$) or disjunction ($+$). In other words, the former uses Eq. (10) to integrate the three dissimilarity values in R , G , and B histograms, while the latter uses the following equation:

$$d_{hc}(Q, T) = \sum_{m=0}^{M-1} |hr_m^{(Q)} - hr_m^{(T)}| + |hg_m^{(Q)} - hg_m^{(T)}| + |hb_m^{(Q)} - hb_m^{(T)}|, \quad (30)$$

where $hr_m^{(Q)}$, $hg_m^{(Q)}$, $hb_m^{(Q)}$ and $hr_m^{(T)}$, $hg_m^{(T)}$, $hb_m^{(T)}$ are the normalized frequencies in bin m of R , G , and B histograms, respectively. We then classify Corel-20 using color histogram with different numbers of bins and different combination strategies.

Quantization results using different number of bins are shown in Fig. 9. The classification results for different options are shown in Fig. 10. From the experimental results, we found that the more bins, the higher the classification accuracy, but the longer the processing time. On the other hand, the conjunction combination ($\times$) has better performance but longer execution time. So, we get two compromise solutions, i.e. $4 \times 4 \times 4$ and $8 \times 8 \times 8$ in this study.

4.1.2. Options related to the LEP histogram

There are two options related to the LEP histogram. The first is whether to obtain the edge image from the quantized image or the original image. The second is whether to use the original or the conditional LEP histograms. The resulting edge images from the quantized and original image are shown in Fig. 11. We then classify Corel-20 in the original LEP histogram using the different

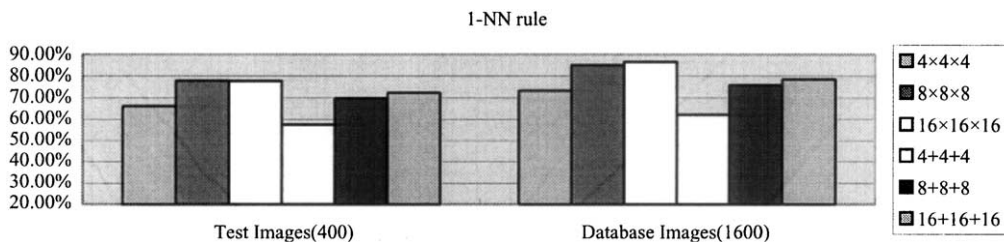


Fig. 10. Comparison of classification accuracies of using different numbers of bins and different combinations.

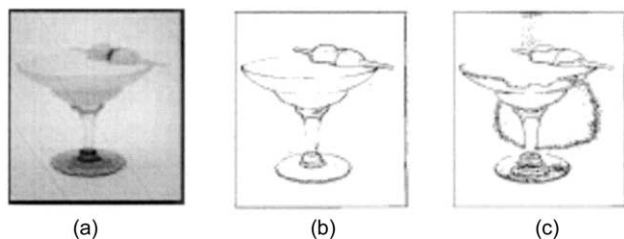


Fig. 11. Comparison of edge images. (a) Original image; (b) edge image from original image; (c) edge image from quantized image (4 bins).

resulting edge images. The classification results for different options are shown in Fig. 12. It is obvious that the LEP histogram from the edge image obtained from the original image leads to good performance, so we decided to use it.

We classify Corel-20 using different LEP histograms as specified by Eqs. (11) and (13). The corresponding classification results are shown in Fig. 13. It is obvious that the conditional LEP histogram performs better, so we decided to use it.

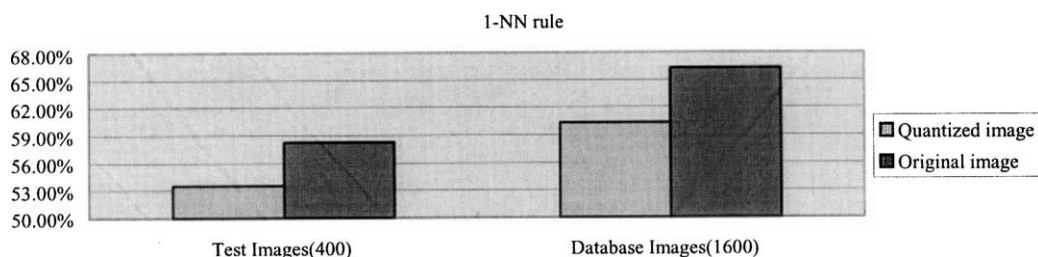


Fig. 12. Comparison of classification accuracies of using edge images obtained differently.

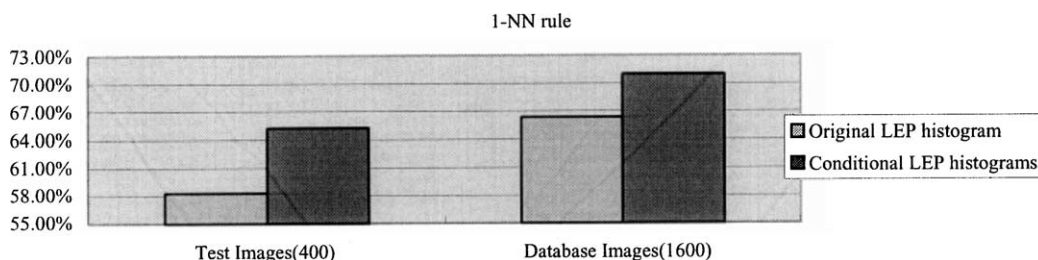


Fig. 13. Comparison of classification accuracies of using different LEP histograms.

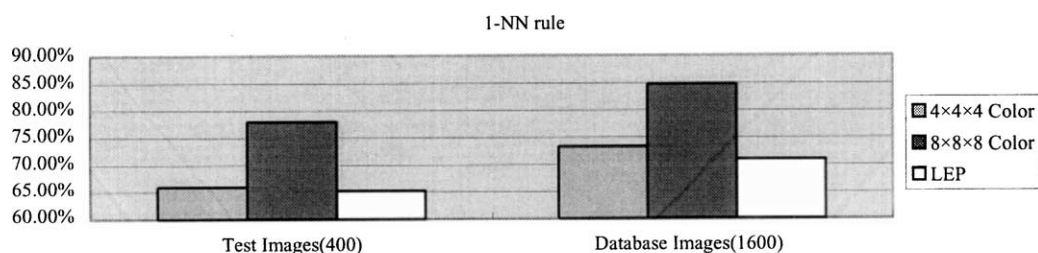


Fig. 14. Comparison of classification accuracies of using color and LEP histograms separately.

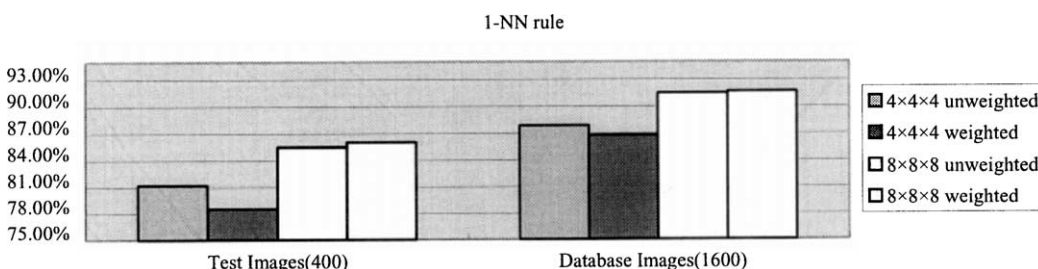


Fig. 15. Comparison of classification accuracies of using different weights.

Region 1	Region 2	Region 3
Region 4	Region 5	Region 6
Region 7	Region 8	Region 9

Fig. 16. Examples of divided blocks.

4.1.3. Integrating image-based features

We first classify Corel-20 using color and LEP histograms separately. The results are shown in Fig. 14. It is obvious that the color histogram outperforms the LEP histogram. We then classify Corel-20 using unweighted and weighted combinations of color and LEP histograms. Note that we try a weighted combination of color and LEP histograms with weights (0.6, 0.4), respectively, since the color histogram has better discriminating power as shown in Fig. 14. However, from Fig. 15, we found that the unweighted combination performs well, particularly for

the $4 \times 4 \times 4$ case. So the image-based dissimilarity is computed by Eq. (14) in this study.

4.1.4. Selection of region-based features

Regions are obtained from segmented images. The image can be segmented adaptively by the segmentation method described in Section 2, or just divided into a fixed number of blocks. The division of an image into 9 fixed blocks is shown in Fig. 16.

For brevity, we use only two region-based features, the region color mean and the region edge ratio, to calculate dissimilarity for each region matching pair. These features are extracted from the fixed blocks or the adaptive regions individually. The results of classifying Corel-20 using different segmentation strategies are shown in Fig. 17. From this figure, we concluded that the region-based features using the adaptive regions perform better. So, region-based features extracted from adaptive regions are adopted for our method.

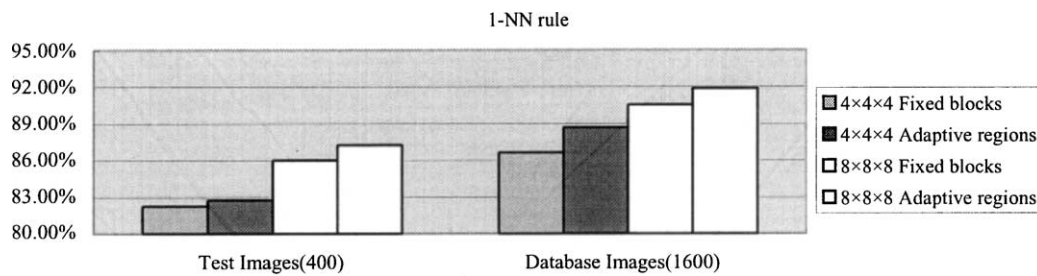


Fig. 17. Comparison of classification accuracies of using fixed blocks and adaptive regions.

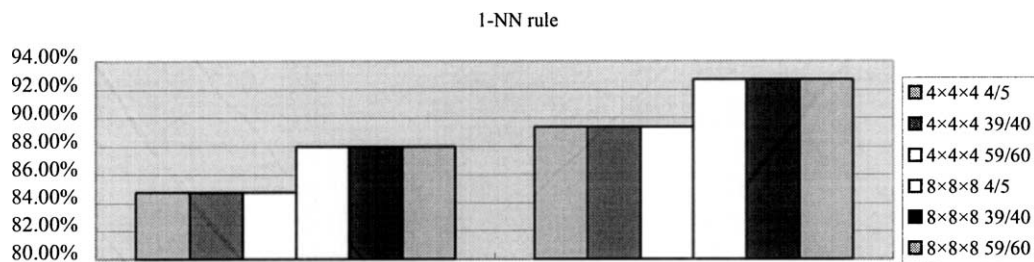


Fig. 18. Comparison of classification accuracies of using different pruning rates.

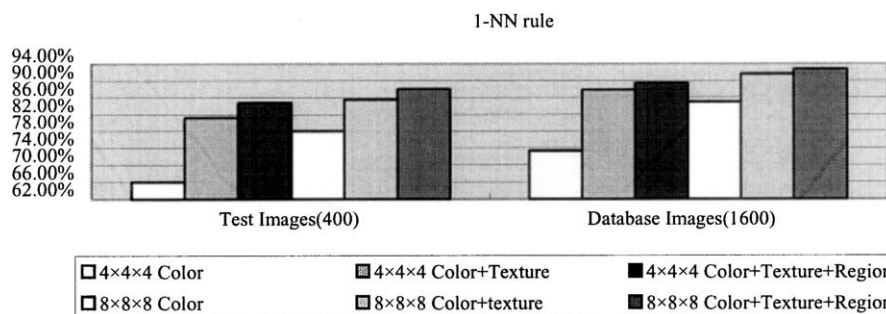


Fig. 19. Comparison of classification accuracies by adding the features color, texture and region step by step.

4.1.5. Determination of pruning rate

As explained in Section 3.3, it is necessary to prune irrelevant images to speed up classification. However, the number of images pruned does affect classification accuracy. The classification results using different pruning rates for Corel-20 are shown in Fig. 18. We

concluded that the 39/40 pruning rate is satisfactory since the case of 39 out of 40 database images getting pruned has the same accuracy as that of 4 out of 5 and 59 out of 60, but the case of 4/5 pruning rate executes slower and the case of 59/60 pruning rate has too few database images remaining. So, the proposed method



(a)



(b)

Fig. 20. Retrieval results by adding features step by step. (a) Only color histogram; (b) only LEP histogram; (c) unweighted combination of color and LEP histograms; (d) our method.



(c)



(d)

Fig. 20 (continued)

adopts the 39/40 pruning rate and keeps at least 40 database images remain.

4.2. Evaluation classification performance

First, we demonstrate that the features of color, texture and region used in this paper are necessary and neither

one can be deleted. We add the features step by step to show that the performance increases gradually. From Fig. 19, we found that the proposed method using all the features, color, texture and region leads to the highest classification accuracy. In addition, a retrieval result as shown in Fig. 20 is included to show that using all the features leading to more correct retrieval results.

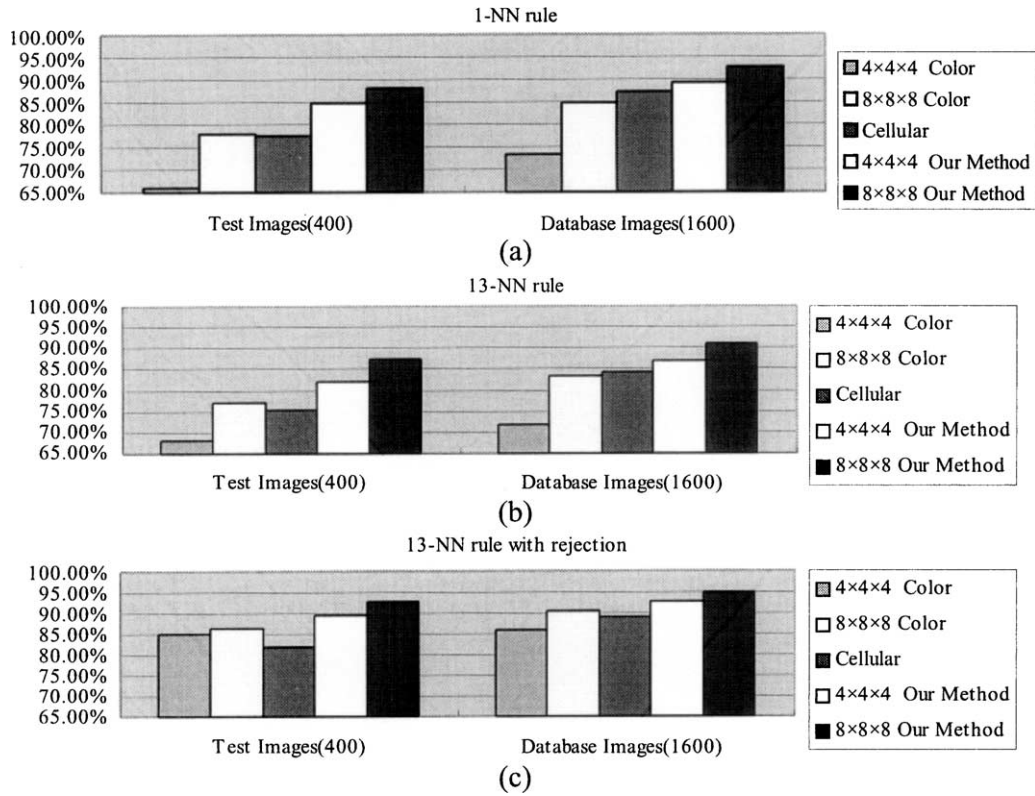


Fig. 21. Comparison of classification accuracies of our method to color histogram intersection and cellular color decomposition for Corel-20. (a) 1-NN rule; (b) 13-NN rule; (c) 13-NN rule with rejection.

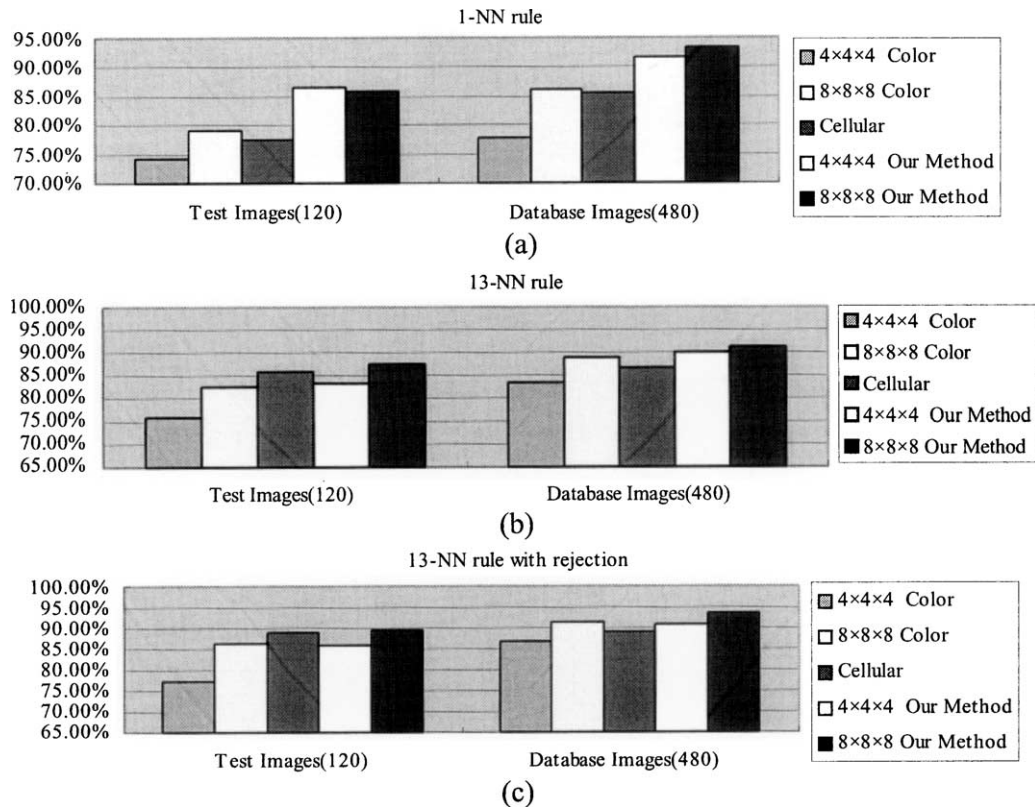


Fig. 22. Comparison of classification accuracies of our method to color histogram intersection and cellular color decomposition for Corel-6. (a) 1-NN rule; (b) 13-NN rule; (c) 13-NN rule with rejection.



Fig. 23. Retrieval result 1 for Corel-20.



Fig. 24. Retrieval result 2 for Corel-6.



Fig. 25. Retrieval result 3 for Corel-6.



Fig. 26. Failed retrieval result 1 for Corel-20.

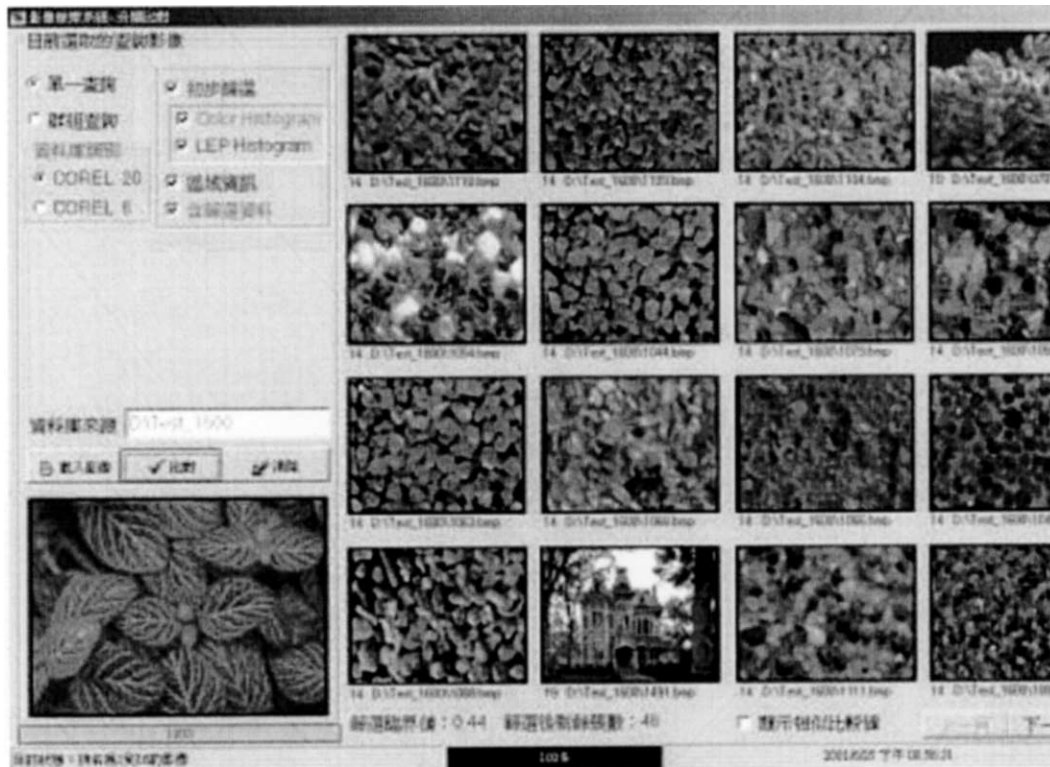


Fig. 27. Failed retrieval result 2 for Corel-20.

We then compare our method to two existing methods. One is color histogram intersection [1], which we denote Color. In our experiment, Color was implemented as a pure color indexing without using region segmentation. The other uses adaptive cellular color decomposition for image classification [12], which we denote Cellular. However, Cellular involves the region information implicitly. In addition, the region denotes a set of contiguous pixels with homogeneous color in Cellular rather than a set of contiguous pixels with homogeneous texture in our method. The method Cellular uses image-based and cluster-based features. The former includes color width, region ratio, edge ratio, edge density variance and initial color histogram. The latter includes pixel occurrence and region occurrence. The comparison results are shown in Figs. 21 and 22, from which we found that our method is most effective. On average, the classification accuracies of 1-NN rule of Corel-20 are 88.45 and 91.75% for $4 \times 4 \times 4$ and $8 \times 8 \times 8$ quantization, and 90.67 and 91.83% for Corel-6.

Some retrieval results are also shown in Figs. 23–27. Some retrieval errors may be caused by the presence of ambiguous images in the database. For example, if a query image contains airplane and cloud, the retrieved images may include some cloud images even though airplane images are required as shown in Fig. 26. On the other hand, images in different classes may have similar textures, yielding retrieval error. For example, an image of foliage background has regular texture like those

images in the class of sand and pebble textures as shown in Fig. 27.

5. Conclusions

In this study we propose a new method for classifying images. From the results as shown in Section 4, we conclude that our method has the following advantages.

- (1) The adopted color texture segmentation method unifies color and texture features to obtain homogeneous regions.
- (2) The proposed region matching method can correctly match regions between two images so that region-based dissimilarity can improve the classification accuracy.
- (3) The proposed classification method is effective, yet simple. On average, the classification accuracies of 1-NN rule for Corel-20 are 88.45 and 91.75% for $4 \times 4 \times 4$ and $8 \times 8 \times 8$ quantization, and 90.67 and 91.83% for Corel-6.

In a summary, the main contribution of this paper is to select proper features for representing color, texture and region, which, in turn, are used to achieve effective classification results. More important, all features used in the proposed method, no matter color or texture, are presented in the simple form of histogram, yet leading to

effective results. Even in the stage of region matching, color and texture features in histograms are also used to obtain homogeneous regions and to measure dissimilarity. In addition, the proposed classification method can be applied to all kinds of color image databases rather than specific databases. The number of classes can be as versatile as required by the application.

Future work can be directed to the following topics. First, subtler feature description must be designed to improve segmentation results. Second, intelligent strategy to find significant regions is needed to increase classification accuracy. Third, more experiments must be conducted to prove the effectiveness and efficiency of the proposed method.

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